PRICING AND PRODUCTION WITHOUT THE INVISIBLE HAND

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October 2026

MOTIVATION

- **Arrow's (1959) Critique**: Classical competitive models have no rational price-setting decision:

  > *"…there is no place for a rational decision with respect to prices as there is with respect to quantities."*

- **The Modern New Keynesian Inversion**:
  - Firms optimize prices, but are forced to produce whatever quantity the market demands at that price.
  - Violates the core economic principle of **voluntary exchange**: firms must produce even in loss-making states.

- **Breakthrough**: A framework where optimizing firms simultaneously choose **both prices and inputs** under uncertainty.

WHY STUDY DISEQUILIBRIUM?

- **The Salon Decision**: The manager sets haircut prices and schedules stylists in advance, before realized customer demand is known.
  - **High Demand (Rationed)**: Salon is full; prospective walk-ins are turned away ($q = q^s < q^d$).
  - **Low Demand (Slack)**: Stylists sit idle; labor capacity is wasted ($q = q^d < q^s$).
  - Capacity equals demand only *"on average"*.

- **Macroeconomic Relevance**:
  - Services represent ≈ 80% of US GDP.
  - Frequent retail stockouts across the business cycle.
  - Pervasive economic slack: USDA estimates 31% of US retail food supply is wasted; substantial idle worker time.

FIRM SETUP: PARTIAL EQUILIBRIUM

- **Technology & Demand**: Monopolist chooses $(p, L) \in \mathbb{R}^2_{++}$ before $(A, z, w, \Lambda) \sim$ Log-$\mathcal{N}$ realize:

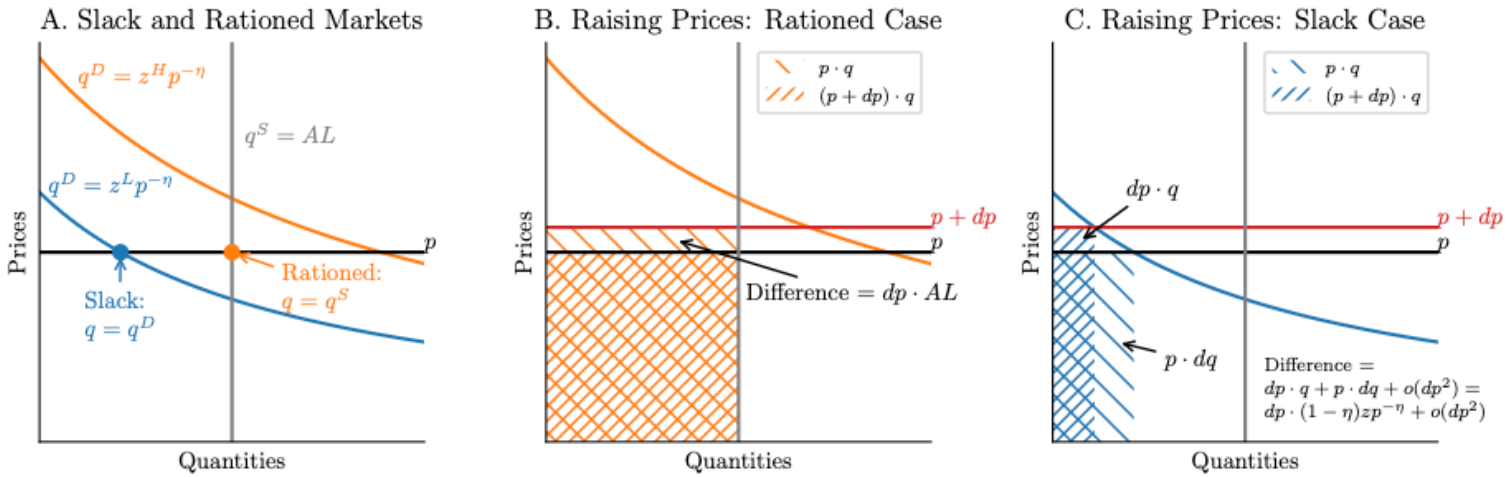
$$q^s = AL \quad \text{and} \quad q^d = zp^{-\eta} \quad (\eta > 1)$$

- **Axioms of Exchange** (Bénassy 1993):
  1. *Voluntary Exchange*: $q \leq q^s$ and $q \leq q^d$ (no forced trade).
  2. *Efficient Exchange*: No unrealized mutual gains

     $\Longrightarrow \; q = \min\{q^s, q^d\}$.
- **Ex-Post Market States**:
  - Slack: $q^s > q^d$ $(t \leq U) \implies q = zp^{-\eta}$
  - Rationed: $q^s < q^d$ $(t \geq U) \implies q = AL$
  - Walrasian Equilibrium ($q^s = q^d$): Probability zero
- The profit function is **kinked** at market clearing (since revenue is!).

FIGURE 1: PRICE-SETTING INCENTIVES ACROSS STATES

Figure 1: The Incentives to Raise Prices



Note: This figure illustrates the incentives for price setting in the model. Panel A illustrates cases where the market is rationed or slack, depending on the realization of demand. Panels B and C show the marginal revenue effect of increasing prices from p to $p + \mathrm{d}p$ in states that are rationed versus slack, respectively. Equation 7 is obtained by setting the expected (risk-adjusted) marginal revenue effect to zero, averaging over slack and rationed cases under the stochastic discount factor, and Equation 5 of Theorem 1 is obtained by rearranging this condition.

OPTIMAL TIGHTNESS: THEOREM 1

- **Market Tightness**: Define $t \equiv p^{-\eta}/L$ and supply-demand shock $U \equiv A/z$.
- **Key Assumption**: Marginal effect of changing prices is state-dependent.
- **Theorem 1**: Optimal tightness t^* solves the unique fixed point:

$$t = \frac{1}{\eta - 1} \frac{\mathbb{E}[\Lambda A \cdot \mathbb{I}[t \geq U]]}{\mathbb{E}[\Lambda z \cdot \mathbb{I}[t \leq U]]}$$

- At the margin, expected gains when rationed equal expected losses when slack; this is a fixed point because rationing probabilities depend on choices solely through t.
- Similar logic follows for optimal pricing condition, where firms are indifferent to marginally changing inputs.

THE MARKUP EQUATION: INVERSE-RATIONING RULE

- Optimal price satisfies:

$$p^* = \frac{1}{r} \frac{\mathbb{E}[\Lambda w]}{\mathbb{E}[\Lambda A \mid t^* \geq U]}, \quad \text{where } r \equiv \mathbb{P}[t^* \geq U]$$

- The markup is the inverse rationing probability, applied to an object close to marginal cost, with two twists: (i) risk adjustment by the SDF Λ, and (ii) a selection correction
- **Deviation from Lerner (1934)**:
  - Demand elasticity η does not appear directly in the markup formula! still matters indirectly, through its effect on t and thus on r.
  - Hiring a marginal worker incurs wage cost w in **all** states, but generates revenue pA **only when rationed**.

THEOREM 2: UNCERTAINTY HAS FIRST-ORDER EFFECTS

- **Theorem 2**: Expanding around the full-information limit ($\delta = 0$):

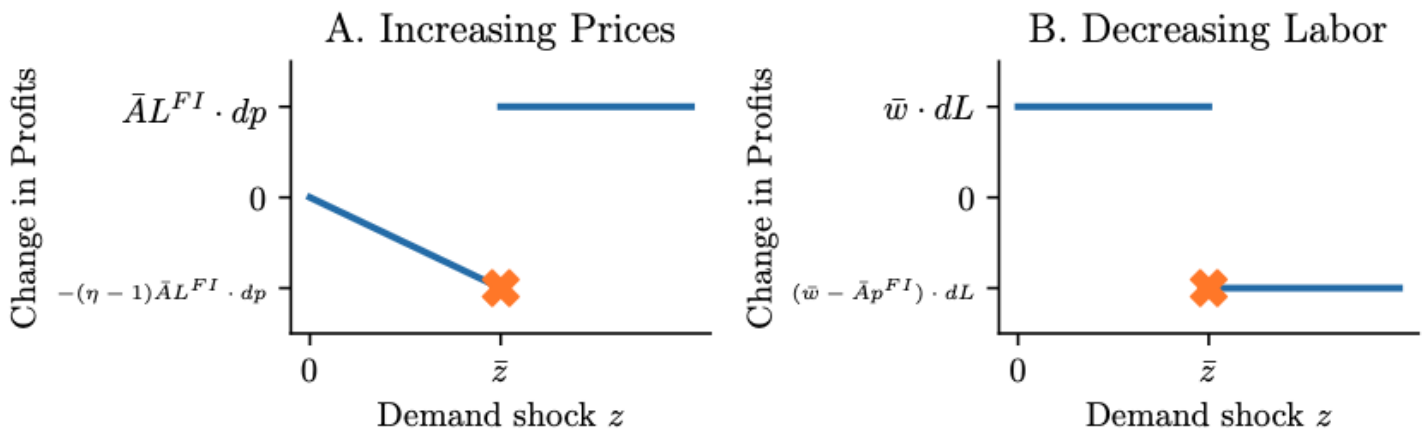
$$\ln p^* = \ln p^{\text{FI}} + \underbrace{\left(\phi(\kappa) - \frac{\kappa}{\eta}\right)}_{\Omega > 0} \sigma_U, \quad \ln L^* = \ln L^{\text{FI}} - \eta\phi(\kappa)\sigma_U$$

  where $\kappa = \Phi^{-1}((\eta - 1)/\eta)$ and $U = A/z$ is the sufficient statistic.

- Unlike standard DSGE where volatility matters only to II or III order (Schmitt-Grohé & Uribe 2004), σ_U enter decisions to **first order**.
- Uncertainty moves in a specific **direction**: a more uncertain firm sets a higher price and hires less labor.
- Expected marginal profit from raising p is strictly positive in both states, while marginal profit from labor is negative.

FIGURE 2: DEMAND SHOCKS AND MARGINAL INCENTIVES

Figure 2: Demand Shocks and Marginal Incentives



Note: This figure shows the marginal effects on profits of increasing prices by dp (panel A) and reducing labor by dL (panel B) as a function of the realized demand shock z, starting from choices (p^{FI}, L^{FI}) that maximize profits conditional on full knowledge that $z = \bar{z}$. The orange X's indicate marginal effects in this full information case. The blue lines correspond to marginal effects conditional on other possible realizations of z.

IMPLICATIONS: ENDOGENOUS MARKUPS & THE KINK

- **Markup Decomposition**:

$$\ln \mathcal{M} = \underbrace{\ln\left(\frac{\eta}{\eta - 1}\right)}_{\text{Taste-Based Markup}} + \underbrace{\Omega \cdot \sigma_U}_{\text{Uncertainty-Based Markup}}$$

- **Payoff 1: Resolves Empirical Markup Puzzle**:
  - Demand-side estimates find markups near 1.1.
  - Supply-side production estimates find markups of $1.20 - 1.60$.
  - The wedge is explained by a strictly positive uncertainty markup.
- **Payoff 2: Microfounds Markup Shocks**: Fluctuations in firm uncertainty act as cost-push shocks without preference shifts.
- **The Kink is Essential**: Proposition 7 proves that smoothing the kink via a matching function kills all first-order volatility effects!

GENERAL EQUILIBRIUM - BLANCHARD AND KIYOTAKI (1987)

- Continuum of firms $i \in [0,1]$, CES varieties, Golosov-Lucas HH preferences ($w = \alpha M$, $PC = M$, $\Lambda = 1/M$).
- Family spends E subject to variety capacity constraints $c_i \leq \bar{c}_i = A_i L_i$.
- **Supply-Demand Imbalance Statistic** ι_i (Proposition 2):

$$\iota_i \equiv \frac{p_i^{-\eta}/L_i}{A_i/\theta_i} = \frac{\theta_i p_i^{-\eta}}{\bar{c}_i} \quad \text{[desired demand relative to available supply]}$$

  There is a unique threshold index $k \in [0,1]$, where the household is exactly indifferent between buying all of it and a little less.
- The **aggregate consumption index** is then computed from this split allocation mixing fully-consumed rationed and demand-curve slack varieties.

GENERAL EQUILIBRIUM - BLANCHARD AND KIYOTAKI (1987)

- **Demand for slack markets** ($c_i \leq \bar{c}_i$):

$$c_j = D\theta_j p_j^{-\eta} \quad \text{where } D = \frac{E - \int_0^k p_i \bar{c}_i \, di}{\int_0^k \theta_i p_i^{1-\eta} \, di}$$

- **Cross-Market Spillovers**: Consumers stock out in rationed markets ($i \in [k, 1]$); their unspent liquidity spills over to slack goods.
- **Negative Aggregate Demand Externality**:
  - Standard literature (Blanchard-Kiyotaki 1987): Production is a strategic complement ($\bar{c}_i \uparrow \implies$ AD $\uparrow$).
  - Here: Expanding output in rationed markets absorbs household spending, depressing effective demand in slack markets!

EQUILIBRIUM: PROPOSITION 3

- **Proposition 3**: Aggregate output satisfies $C = \mathcal{A}L$, where measured aggregate labor productivity $\mathcal{A}$ decomposes into:

$$\mathcal{A} = \frac{\eta}{\eta - 1} \times \underbrace{\frac{p}{P}}_{\text{Misallocation}} \times \underbrace{r}_{\text{Rationing}} \times \underbrace{\mathbb{E}[A_i \mid t \geq U_i]}_{\text{Physical Productivity}}$$

- **Three Wedges in the Solow Residual**:
  1. **Price Dispersion Wedge** ($p/P < 1$): Rationing distorts consumption deflator P away from posted prices p.
  2. **Rationing Fraction** (r): Higher rationing means less unsold output is wasted in slack markets.
  3. **Selection in Physical TFP**: Only productivity in supply-constrained (rationed) markets matters on the margin.
- Shocks move the macroeconomy via $\mathcal{A}$, holding L constant.

MACRO SHOCKS: THEOREMS 3 AND 4

- **Uncertainty Shocks (Theorem 3)**:

$$\ln P = \ln P^{*,\mathrm{FI}} + \Omega\sigma_U, \quad \ln C = \ln C^{*,\mathrm{FI}} - \Omega\sigma_U$$

  - Acts as an endogenous **stagflationary cost-push shock**.
  - Propagates through collapsing aggregate productivity ($d\ln\mathcal{A} = -\Omega d\sigma_U$), microfounding Smets-Wouters shocks.

- **Monetary Shocks (Theorem 4)**:
  - Unanticipated monetary expansion ($M\uparrow$) raises consumption **purely by absorbing slack** with predetermined p and L.
  - **Discontinuous Passthrough**: Fully neutral in zero-uncertainty limit ($\mathcal{H}^+(0) = 0$ vs. $\mathcal{H}^-(0) = 1$) because slack vanishes.
  - Passthrough $\mathcal{H}$ depends non-monotonically on competition (minimized at $\eta^* \approx 2.83$).

DYNAMICS: THE NEW OLD KEYNESIAN (NOK) MODEL

- **Setup**: Calvo (1983) staggering on **both prices and inputs** (p and L).
- **Log-Linearized 3-Equation System** (Proposition 6):
  1. **New Keynesian Phillips Curve**:

$$\pi_\tau = \Gamma(\hat{M}_\tau - \bar{p}_\tau) + \beta \mathbb{E}_\tau[\pi_{\tau+1}] + \zeta \hat{\sigma}_{U,\tau}$$

  2. **NOK Tightness Curve**:

$$\upsilon_\tau = -\Gamma(\hat{D}_\tau + \bar{t}_\tau) + \beta \mathbb{E}_\tau[\upsilon_{\tau+1}] + \psi \hat{\sigma}_{U,\tau}$$

  3. **Demand Spillovers Curve**:

$$\hat{D}_\tau = \gamma_M \hat{M}_\tau + \gamma_p \bar{p}_\tau + \gamma_t \bar{t}_\tau + \gamma_U \hat{\sigma}_{U,\tau}$$

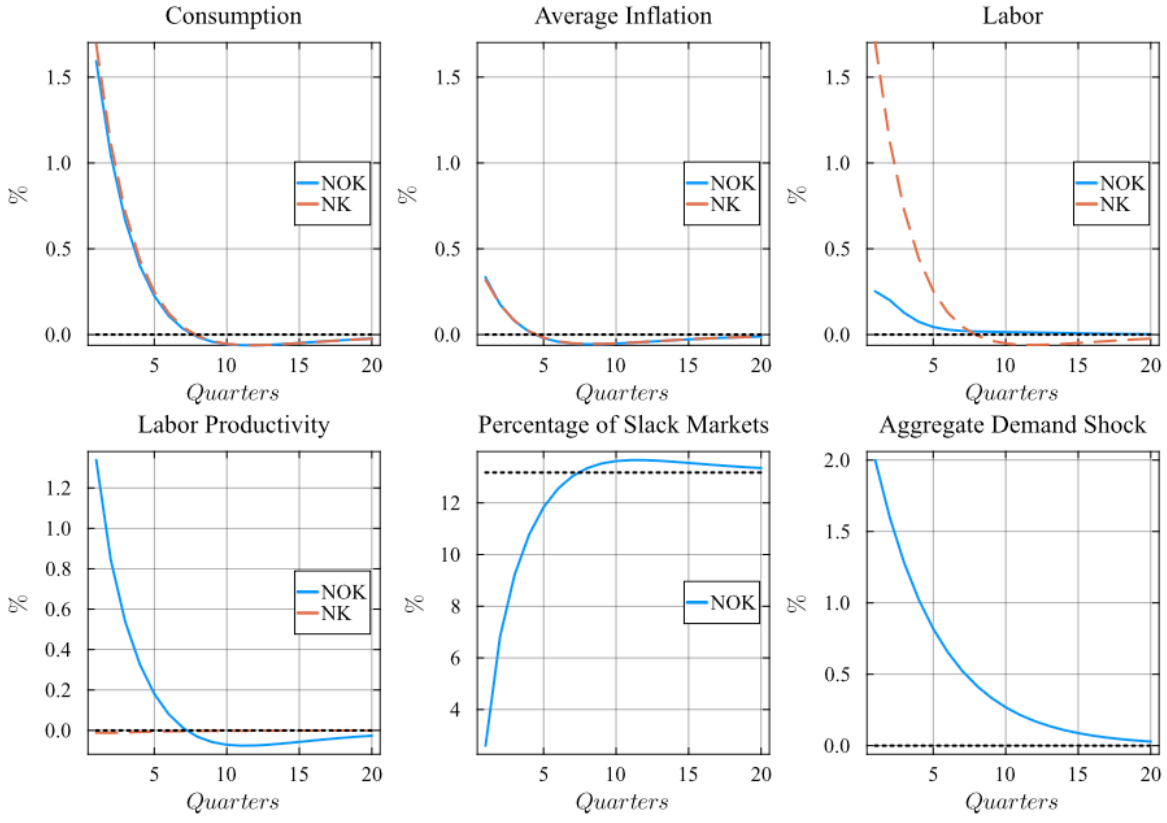
- Inflation dynamics are identical to standard NK to first order, but physical output and employment dynamics are sharply different!

12. CALIBRATION RESULTS

- **Calibration**: $\beta = 0.99$, $\eta = 8$, Calvo $\omega = 0.75$; micro-calibrated $\sigma_U = 0.23$ ($\sigma_A = 0.10$, $\sigma_z = 0.21$) using US Census TFPQ/TFPR microdata.
- **Monetary Expansion (Figure 4,** $+2\%$ **Demand)**:
  - Consumption surges as slack markets drop from 12.5% to 1%.
  - Employment barely moves on impact; labor productivity surges.
  - Matches Buda et al. (2025) high-frequency evidence (sales move $\approx 1\%$, employment only $\approx 0.1\%$) and Christiano et al. (2005).
- **Uncertainty Shock (Figure 5,** σ_U **up** $+43\%$**)**:
  - Sharp drop in consumption, inflation spike, collapse in labor productivity.
  - In standard New Keynesian model, volatility has **exactly zero effect**!

FIGURE 4: DYNAMIC MONETARY EXPANSION (NOK VS. NK)

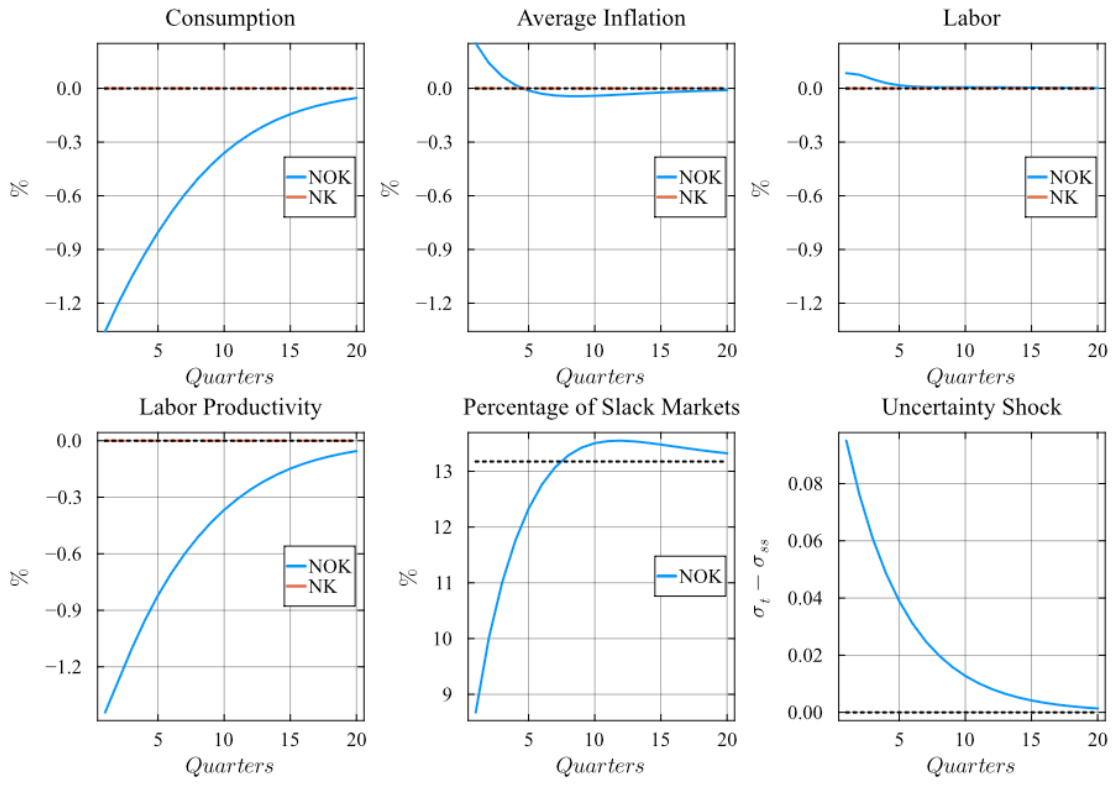
Figure 4: Impulse Responses to an Aggregate Demand Shock



Note: This figure shows the response of macroeconomic variables to a temporary 2% increase in the money supply at $\tau = 0$, beginning from a steady state and decaying at 20% per quarter. Each panel contrasts the responses in the New Old Keynesian model (blue solid line) with those in the New Keynesian model (orange dashed line) with the same parametrization. The bottom right plot shows the path of the money supply M_τ. "Consumption" is the consumption quantity index (Equation 44); "Average Inflation" is the growth in the ideal price index, P_τ; "Labor" is total labor hours; "Labor productivity" is GDP (Consumption) per labor hour; and the "Percentage of Slack Markets" is the percentage of markets for intermediate goods in which supply exceeds demand. All variables except inflation and the percentage of slack markets are expressed in percent deviations from the steady state.

FIGURE 5: DYNAMIC UNCERTAINTY SHOCK (NOK VS. NK)

Figure 5: Impulse Responses to an Uncertainty Shock



Note: This figure shows the response of macroeconomic variables to a transitory increase in uncertainty, beginning from a steady state and decaying at 20% per quarter. Each panel contrasts the responses in the New Old Keynesian model (blue solid line) with those in the New Keynesian model (orange dashed line) with the same parametrization. The shock, shown in the bottom right panel, corresponds to a temporary increase in uncertainty about the idiosyncratic demand shock θ and the idiosyncratic productivity shock A, such that $\sigma_U = \text{StdDev}[\log A - \log \theta]$ rises by 0.1 on impact, and the ratio of uncertainty regarding θ and A remains constant. "Consumption" is the consumption quantity index (Equation 44); "Average Inflation" is the growth in the ideal price index, P_τ; "Labor" is total labor hours; "Labor productivity" is GDP (Consumption) per labor hour; and the "Percentage of Slack Markets" is the percentage of markets for intermediate goods in which supply exceeds demand. All variables except inflation, the percentage of slack markets, and uncertainty are expressed in percent deviations from steady state.

TAKEAWAYS

- **Three Empirical Facts Solved by NOK**:
  1. **Stagflationary Uncertainty**: Spikes in volatility raise prices and depress output without preference shocks.
  2. **Procyclical Productivity**: Labor productivity rises following monetary stimulus without technology shifts.
  3. **Short Sales Lags**: Consumption responds much faster than labor to aggregate demand shocks.

- **Extensions & Modularity**:
  - Appendix extends model to inventories and labor-market disequilibrium.
  - Modular framework: can be plugged directly into large-scale DSGE and open-economy international macro models.